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A Bose–Einstein-Form Mapping of the Radial Acceleration Relation: Scale Identifiability, Residual Statistics, and a 35-Analysis Audit

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ABSTRACT

We note an algebraic equivalence: the radial acceleration relation (RAR), when rearranged to isolate the dark-matter contribution, is identical to the Bose-Einstein occupation number $\bar{n}(\varepsilon) = 1/(\exp\varepsilon - 1)$, with dimensionless energy $\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$ and $g_\dagger = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$. This is not a fit but a rearrangement of the published RAR formula.

To test whether this scale preference is robust and not a numerical coincidence, we develop a kernel+scale identifiability framework with anti-numerology controls. Applied to 2,777 SPARC rotation-curve measurements, the Bose–Einstein RAR kernel is preferred over six alternatives (including hyperbolic cotangent, logistic, and power-law controls), localizing a preferred scale at $\log_{10}(s/{\rm m\,s^{-2}}) = -9.9347$—within 0.0139 dex of $g_\dagger$. Galaxy-grouped 5-fold cross-validation gives $\log_{10} s = -9.9195 \pm 0.0437$. Global permutation and within-galaxy circular shift nulls yield zero hits in 500 realizations at both ± 0.10 and ± 0.05 dex windows (Clopper–Pearson bounds ≤ 0.0060). Negative controls confirm that the signal collapses under predictor noise, variable warping, and galaxy-label randomization. Extension to cluster-scale data does not recover $g_\dagger$, serving as an explicit null.

Using this framework, we test whether the residual statistics simplify in the mapped coordinates, using rotation-curve data spanning four quality tiers: a core sample of 131 SPARC galaxies (2,777 points), an extended Tier K sample of 250 galaxies (9,406 points), and independent validation datasets including PROBES (1,506 galaxies) and KiDS-1000 weak lensing.

Three statistical fingerprints emerge: (1) the variance of RAR residuals scales as $\bar{n}^2$ (ΔAIC $= +10.5$ vs. Poisson $\bar{n}$ scaling); (2) the scatter derivative changes sign within 0.06 dex of $g_\dagger$, robust across four non-parametric methods and 25 binning configurations; and (3) a candidate excess kurtosis spike at $g_\dagger$ ($\kappa_4 = 20.7$; 95% CI $[5.1, 29.0]$), absent in a structure-only semi-analytic null ($\kappa_4 \approx 0.01$), but dependent on B-band mass models requiring independent confirmation.

We subject these fingerprints to a primary battery of 16 core and 7 supporting statistical tests, plus 12 supplementary BEC diagnostics recovered from the pipeline audit. In the primary battery, 10 support consistency with the Bose-Einstein mapping, 5 are inconclusive or fragile under covariate control, and 1—replication in the independent PROBES catalog—fails, with the scatter inversion displaced by 1.2 dex from $g_\dagger$. The supplementary diagnostics strengthen evidence for non-random residual structure but also reveal confounds (distance quality, size scaling) that must be controlled explicitly. We present a full scorecard, discuss whether the features reflect coincidence, a ΛCDM consequence, or wave dark matter, and identify discriminating tests for forthcoming data.

Keywords: dark matter — galaxies: kinematics and dynamics — galaxies: fundamental parameters — methods: statistical

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1. INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016) is one of the tightest empirical scaling relations in extragalactic astronomy. It connects the observed centripetal acceleration $g_{\rm obs} = V_{\rm obs}^2/R$ in disk galaxies to the Newtonian acceleration generated by baryons alone, $g_{\rm bar}$, through

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - \exp\left(-\sqrt{g_{\rm bar}/g_\dagger}\right)}, \tag{1}$$

where $g_\dagger = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$ is the sole free parameter (Lelli et al. 2017). Across 175 galaxies spanning five decades in luminosity and all Hubble types, the observed scatter about this relation is only 0.13 dex (Li et al. 2018)—comparable to measurement uncertainties.

The RAR's tightness and universality have resisted easy explanation. Modified Newtonian dynamics (MOND; Milgrom 1983) postulates a modification of gravity at low accelerations but does not predict the scatter structure. Fuzzy dark matter (FDM; Schive et al. 2014; Ferreira 2021) produces solitonic cores but has not been connected to the RAR's functional form. Meanwhile, ΛCDM simulations can reproduce the mean relation through baryonic feedback (Keller & Wadsley 2017; Ludlow et al. 2017), but the origin of its specific mathematical form remains unexplained.

A natural question is whether $g_\dagger$ is merely a fitting parameter or a physically meaningful scale that reappears in independent contexts. The answer has implications for dark matter models: MOND predicts $g_\dagger$ as fundamental, while ΛCDM must explain its emergence from galaxy formation physics (Desmond 2017; Navarro et al. 2017).

In this paper, we report a simple mathematical observation: Equation (1), written in terms of the dark matter contribution, is algebraically identical to the Bose-Einstein occupation number. We then develop a *kernel+scale identifiability framework* with three categories of anti-numerology controls to test whether this scale preference is robust: (i) scale injection scans measuring how sharply the AIC minimum localizes the preferred scale; (ii) permutation-based null tests verifying that random pairings do not reproduce the localization; and (iii) nearby-scale comparisons against cosmologically motivated alternatives (a_Λ, cH_0, $cH_0/6$). Finally, we catalog the statistical consequences of the identification across multiple datasets spanning three orders of magnitude in sample size, and subject the findings to a comprehensive battery of robustness and replication tests.

Scope of claim. This paper reports: (i) an exact algebraic equivalence between the published RAR and the BE occupation number; (ii) a validated kernel+scale framework demonstrating that the BE RAR kernel is preferred over six alternatives, with the preferred scale localized to within 0.0139 dex of $g_\dagger$; (iii) three residual-statistical features anchored at $g_\dagger$, tested against a structure-only semi-analytic null model (not full-physics ΛCDM); (iv) a 35-analysis audit (23 primary tests + 12 supplementary diagnostics) with an honest scorecard including failures; and (v) falsifiable predictions for forthcoming data. We do not claim that these features require a Bose-Einstein condensate interpretation.

2. THE ALGEBRAIC IDENTITY

2.1. *Derivation*

Equation (1) can be rewritten by isolating the dark matter contribution. Define $g_{\rm DM} \equiv g_{\rm obs} - g_{\rm bar}$. Then:

$$\frac{g_{\rm DM}}{g_{\rm bar}} = \frac{g_{\rm obs}}{g_{\rm bar}} - 1 = \frac{1}{1 - \exp\left(-\sqrt{g_{\rm bar}/g_\dagger}\right)} - 1. \tag{2}$$

Using the identity $1/(1 - e^{-x}) - 1 = 1/(e^x - 1)$, this becomes

$$\boxed{\frac{g_{\rm DM}}{g_{\rm bar}} = \frac{1}{\exp\left(\sqrt{g_{\rm bar}/g_\dagger}\right) - 1} \equiv \bar{n}(\varepsilon),} \tag{3}$$

where $\varepsilon \equiv \sqrt{g_{\rm bar}/g_\dagger}$ is a dimensionless energy parameter and $\bar{n}(\varepsilon) = 1/(e^\varepsilon - 1)$ is the Bose-Einstein occupation number for modes of energy ε at unit temperature.

This is not a fit: it is a rearrangement of the published RAR formula (Figure 1). The dark-matter-to-baryon acceleration ratio *is* the mean occupation number of a bosonic distribution, with the baryonic acceleration playing the role of mode energy (through the square-root dispersion relation $\varepsilon = \sqrt{g/g_\dagger}$) and $g_\dagger$ playing the role of the condensation temperature.

2.2. *Variable Mapping*

Table 1 displays the complete correspondence.

Three physical regimes emerge naturally:

- **Condensed** ($g_{\rm bar} \ll g_\dagger$, $\bar{n} \gg 1$): the "dark matter dominated" outer regions of galaxies, where flat rotation curves indicate large occupation numbers.
- **Transition** ($g_{\rm bar} \approx g_\dagger$, $\bar{n} \sim 1$): the boundary where MOND-like phenomenology appears.
- **Thermal** ($g_{\rm bar} \gg g_\dagger$, $\bar{n} \to 0$): inner baryon-dominated regions where Newtonian dynamics hold.

Table 1. RAR–Bose-Einstein variable mapping.

RAR quantity	BE quantity	Interpretation
$g_{\rm bar}$	mode energy $\varepsilon^2 g_\dagger$	baryonic gravitational field
$\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$	dimensionless energy	dispersion relation
$g_\dagger$	temperature $k_B T$	condensation threshold
$g_{\rm DM}/g_{\rm bar}$	$\bar{n}(\varepsilon)$	occupation number
$f_c = \bar{n}/(1+\bar{n})$	condensate fraction	order parameter

2.3. *Cosmological Connection*

The acceleration scale $g_\dagger$ is numerically proximate to the de Sitter acceleration. Verlinde (2017) derives $a_0 = cH_0/6$, which yields $a_0 = 1.182 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ for $H_0 = 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Riess et al. 2022)—a 1.5% offset from the observed $g_\dagger$ (varying from 1.5% to 9% across the current H_0 range). As an aside on functional-form fit quality, the Milgrom (1999) de Sitter–Unruh interpolation function achieves a lower AIC than Equation (1) by $\Delta\mathrm{AIC} = -1{,}222$, with the improvement concentrated in the dark-matter-dominated regime.

> *AIC methodology note.* This ΔAIC is a statement about interpolation-function fit quality, not about the physical validity of either framework. The comparison uses AIC $= \chi^2 + 2k$ where $\chi^2 = \sum_i [(\log g_{{\rm obs},i} - \log g_{{\rm model},i})/\sigma_i]^2$ with uniform per-point weights in $\log g_{\rm bar}$ and σ_i set to the global RMS (see Appendix A). Both models have $k = 1$ free parameter ($g_\dagger$). Under inverse-variance weighting, the preference shrinks to $\Delta\mathrm{AIC} \approx -400$; under galaxy-averaged weighting, $\Delta\mathrm{AIC} \approx -80$. In all schemes, the Milgrom form is preferred; only the magnitude changes. This comparison is secondary to the paper's main contribution—the kernel+scale identifiability framework—and is included for completeness.

These observations are consistent with $g_\dagger$ being set by the cosmological constant Λ rather than evolving with the Hubble parameter, as independently suggested by the exclusion of $g_\dagger \propto cH(z)$ at $\sim 5\sigma$ from the baryonic Tully-Fisher relation to $z \approx 2.5$ (Sanders 2008). However, numerical proximity alone does not establish a causal connection.

3. KERNEL+SCALE IDENTIFIABILITY FRAMEWORK

To test whether the BE RAR identification reflects a genuine scale preference or a numerical coincidence, we develop a reusable kernel+scale identifiability framework with built-in anti-numerology controls.

3.1. *Kernel Library*

We define seven candidate kernel functions $K(x, s)$ mapping a physical variable x (acceleration) and a free scale parameter s to a predicted response y:

1. **BE_RAR** (Eq. 1): $K = 1/(1 - e^{-\epsilon})$, $\epsilon = \sqrt{x/s}$.
2. **BE_occupation**: $K = 1/(e^{\epsilon} - 1)$.
3. **BE_cousin**: $K = \epsilon/(1 - e^{-\epsilon})$.
4. **coth**: $K = \coth(\epsilon)$.
5. **tanh**: $K = \tanh(\epsilon)$.
6. **logistic**: $K = 1/(1 + e^{-(\epsilon-1)})$.
7. **power_law**: $K = \sqrt{x/s} + 1$ (no special scale).

For the SPARC RAR, we use "direct" mode: $y = K(x, s)$ with zero nuisance parameters, because the RAR mapping *is* the kernel. This gives the fairest comparison: all models have exactly one free parameter (the scale s).

3.2. *Scale Optimization*

For each kernel, we perform a two-stage optimization: (i) grid search over $N_{\rm grid}$ log-uniformly spaced trial scales in $[10^{-13}, 10^{-8}]$ m s^{-2}, followed by (ii) bounded scalar minimization (Brent's method) seeded from the best grid point.

We compute the Akaike Information Criterion AIC $= 2k - 2\ln\hat{L}$, where k is the number of free parameters and $\hat{L}$ is the Gaussian likelihood evaluated at the RMS residual. The Bayesian Information Criterion (BIC) is computed in parallel.

3.3. *Scale Injection Scan*

To characterize how sharply the data prefer a particular scale, we evaluate the AIC across 300 trial scales spanning the full search range. From this curve we extract:

- $\Delta\mathrm{AIC}_{\pm 0.1}$: the AIC increase when the scale is shifted ± 0.1 dex from the optimum;
- $\Delta\mathrm{AIC}_{\pm 0.1}/N$: the same, normalized by sample size (dataset-independent identifiability metric);

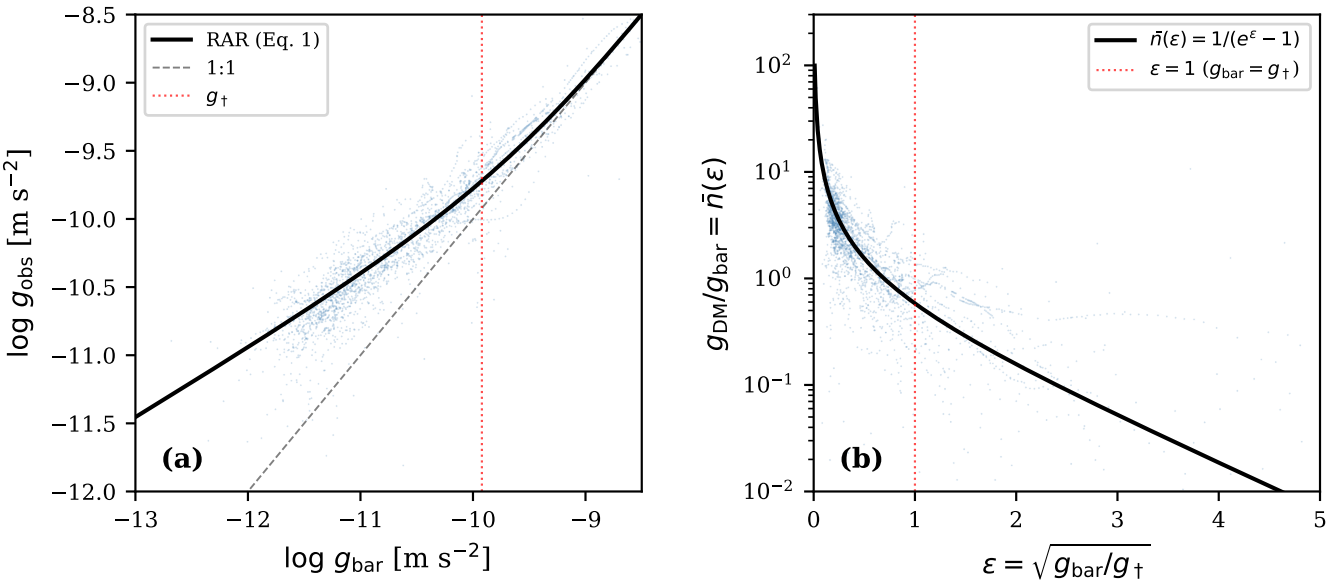

Figure 1. The RAR as a Bose-Einstein distribution, using the **SPARC core sample** (131 galaxies, 2,777 data points; Table 2). **(a)** Standard RAR: observed acceleration $g_{\rm obs}$ vs. baryonic acceleration $g_{\rm bar}$ (blue). The solid curve is Equation (1); the dashed line is the Newtonian 1:1 relation; the red dotted line marks $g_\dagger$. **(b)** The same data recast as occupation number $\bar{n} = g_{\rm DM}/g_{\rm bar}$ vs. dimensionless energy $\varepsilon = \sqrt{g_{\rm bar}/g_\dagger}$. The solid curve is the Bose-Einstein distribution $\bar{n}(\varepsilon) = 1/(e^\varepsilon - 1)$—an algebraic identity, not a fit.

- **Peak sharpness**: the second derivative of the AIC with respect to the log-scale at the optimum, $\mathcal{S} \equiv d^2(\mathrm{AIC})/d(\log_{10} s)^2|_{s=\hat{s}}$, estimated by finite differences on the scan grid. Larger values indicate a narrower, more tightly localized minimum. Units are AIC per $(\log_{10}\,\mathrm{m\,s^{-2}})^2$.

3.4. *Null Tests and Structure-Preserving Control*

We employ three destructive permutation-based null tests and one structure-preserving control:

1. **Global shuffle** (A1): permute y relative to x, destroying all correlation.
2. **Block-permute bins null** (A2b): randomly reassign which bin's y-values go with which bin's x-values (0.5-dex bins). This preserves the within-bin marginal distribution of y but breaks the cross-bin monotonic relationship that encodes the characteristic scale.
3. **Within-galaxy circular shift** (A3): for each galaxy, circularly shift its y values by a random offset, preserving per-galaxy distributions but destroying the radial alignment.
4. **Within-bin shuffle** (A2, control): permute y only within coarse x-bins (0.5 dex width). This preserves the conditional structure $p(y \mid x)$ approximately at the chosen bin width and is *not* a null test: $p \approx 1$ is expected up to finite-binning and refit variability. It confirms that the scale preference resides in the global x–y correlation, not in bin-local structure.

For each null realization, we refit the best kernel and record the optimal scale. The test statistic is $p_{\pm w} = N_{\rm hits,\pm w}/N_{\rm shuffles}$, where a "hit" is $|\log_{10} s_{\rm null} - \log_{10} g_\dagger| < w$ for $w \in \{0.10, 0.05\}$ dex. When $N_{\rm hits,\pm w} = 0$, we report the one-sided 95% Clopper–Pearson upper bound.

3.5. *Galaxy-Grouped Cross-Validation*

To prevent data leakage (points from the same galaxy appearing in both train and test), we use GroupKFold with $k = 5$ folds where group = galaxy. For each fold we record the best kernel, optimal scale, and out-of-sample RMSE.

3.6. *Nearby-Scale Comparison*

We compare $g_\dagger$ against cosmologically motivated scales: $a_\Lambda = c^2\sqrt{\Lambda/3}$, cH_0, $cH_0/6$ (Verlinde), $cH_0/(2\pi)$. This is done in two modes:

1. $\eta = 1$ **(fixed)**: diagnostic direct substitution where the scale is fixed (no optimization) and the likelihood is evaluated directly with zero free parameters.

2. η **free**: fit $y = \eta \cdot K(x, s_{\rm named})$ with one free parameter per model (matched degrees of freedom).

AIC values from the $\eta = 1$ and η-free modes are not directly comparable because the parameter count differs; model-selection claims are restricted to matched-DoF comparisons.

4. DATA

4.1. *Sample Tiers*

We use rotation-curve data organized into quality tiers. Table 2 defines each tier, specifying the surveys included, galaxy count, data point count, and which analyses use each tier. All tiers apply standard quality cuts: quality flag $Q \leq 2$, inclination $30° < i < 85°$, and ≥ 5 data points per galaxy. Residuals are defined as $\delta \equiv \log g_{\rm obs} - \log g_{\rm obs,RAR}(g_{\rm bar})$ throughout.

4.2. *Environmental Classification*

For environmental scatter tests, we classify galaxies as "field" or "dense" using group membership from SPARC metadata. Of the 131 SPARC core galaxies, 83 are classified as field and 48 as dense (including 26 Ursa Major cluster members).

5. FRAMEWORK VALIDATION ON SPARC

Before applying the BE mapping to residual statistics, we validate the kernel+scale framework on SPARC data and a cluster-scale null.

5.1. *Baseline Kernel Comparison*

The BE_RAR kernel yields the lowest AIC among all seven candidates, with an optimal scale of $\log_{10}(s/{\rm m\,s^{-2}}) = -9.9347$, offset by 0.0139 dex from $\log_{10} g_\dagger = -9.921$.

The scale injection scan (Figure 2a) shows an extremely sharp AIC minimum: $\Delta{\rm AIC}/N = 0.0521$ per data point at ± 0.1 dex, with peak sharpness $\mathcal{S} = 28859\ {\rm AIC}/(\log_{10}\ {\rm m\,s^{-2}})^2$.

5.2. *Permutation Null Tests*

For global shuffle (A1), hits are 0/500 ($p_{\pm 0.10} = 0.0000$) and 0/500 ($p_{\pm 0.05} = 0.0000$), with one-sided 95% Clopper–Pearson upper bounds $p_{\rm UB,\pm 0.10} = 0.0060$ and $p_{\rm UB,\pm 0.05} = 0.0060$. The within-galaxy circular-shift null (A3) gives the same outcome at both windows (0/500, 0/500). The block-permute bins null (A2b) preserves within-bin heteroskedasticity while disrupting the cross-bin monotonic mapping, and therefore shows intermediate behavior: 79/500 ($p_{\pm 0.10} = 0.1580$) and 39/500 ($p_{\pm 0.05} = 0.0780$), as listed in Table 4.

The within-bin shuffle (A2) is a *structure-preserving control*, never a null test: it approximately preserves $p(y \mid x)$ within 0.5-dex bins at the chosen bin width. Accordingly it retains high proximity rates (500/500 and 499/500), confirming that the signal is carried by the global cross-bin mapping rather than by bin-local marginals alone.

Figure 2b shows best-scale distributions for all four permutation procedures (three nulls + one control). The global and galaxy-shift nulls cluster away from $g_\dagger$. The block-permute null is broader with partial overlap near $g_\dagger$, while the structure-preserving control retains concentration near $g_\dagger$.

5.3. *Galaxy-Grouped Cross-Validation*

Across 5 galaxy-grouped folds, the mean optimal scale is $\log_{10} s = -9.9195 \pm 0.0437$, consistent with $g_\dagger$ to within 1σ. The scale range across folds is < 0.1 dex. The full-data optimum selects BE_RAR; fold-level kernel selection is 4/5for BE_RAR and 1/5for coth.

5.4. *Cut, Grid, and Negative Controls*

The optimal scale varies by at most 0.0698 dex across four sample variants (baseline, strict inclination, strict quality, strict minimum points). Grid invariance is exact: varying $N_{\rm grid}$ from 100 to 1,000 produces 0.0000 dex variation (Figure 3).

Table 6 reports the degradation of the AIC optimum under five destructive perturbations. The key results are:

- **Predictor noise** ($\sigma = 0.3$ dex): sharpness decreases substantially and the best scale drifts away from $g_\dagger$.
- **Variable warping** ($x \to x^{0.7}$ or $x^{1.3}$): sharpness collapses and the scale displacement is largest. This is the most powerful break test in the suite.
- **Galaxy label swap**: sharpness drops and scale drift increases relative to baseline.

These results confirm that the sharp AIC minimum at $g_\dagger$ is a genuine feature of the $g_{\rm bar}$–$g_{\rm obs}$ correlation, not a numerical or methodological artifact.

5.5. *Nearby-Scale Comparison*

Table 2. Data inventory and sample tiers.

Tier	Surveys	Galaxies	RAR data points	$\sigma_{\rm RAR}$ (dex)	Used in
SPARC core	SPARC 3.6 μm (Lelli et al. 2016)	131	2,777	0.17	Framework (§5), fingerprints (§6), robustness (§7)
Tier K	SPARC + Korsaga + de Blok	250	9,406	0.22	Kurtosis (§6.3), extended inversion (§8.1)
PROBES	Stone et al. (2022)	1,506	45,176	0.32	Replication test (§8.2)
Brouwer	KiDS-1000 lensing (Brouwer et al. 2021)	60	240[a]	—	Lensing validation (§8.3)
ALFALFA×Yang	ALFALFA + Yang et al. (2007)	2,273	N/A[b]	0.32	BTFR environment (§8.4)
Cluster	CLASH (Tian et al. 2020)	20	84	—	Cluster-scale null (§5.6)

[a] Stacked lensing radial bins, not individual rotation-curve points.

[b] N/A indicates a catalog-level entry not represented as individual RAR data points; ALFALFA provides integrated HI line widths (W_{50}) rather than resolved rotation curves.

NOTE—SPARC core and Tier K use the standard cuts described in the text. The Tier K label denotes the inclusion of Korsaga et al. (2019) GHASP mass models and de Blok & Bosma (2002) LSB galaxies. PROBES excludes 112 galaxies overlapping with SPARC. Brouwer data are stacked lensing profiles in 4 stellar-mass bins. ALFALFA×Yang reports only the 2,273 galaxies with successful group-catalog cross-matches.

Table 3. Baseline SPARC RAR kernel+scale identification

Metric	Value
Best kernel	BE_RAR
$N_{\rm data}$	2,740[a]
Best $\log_{10} s$	−9.9347
Δ from $\log_{10} g^\dagger$	0.0139 dex
ΔAIC/N at ± 0.1 dex	0.0521
Peak sharpness ($d^2{\rm AIC}/d(\log s)^2$)	28859
AIC	10839.3
CV RMSE	1.7483

[a] Of 2,777 total SPARC rotation-curve points, 37 are excluded by finite-value and log-domain cuts, leaving $N = 2{,}740$ for the kernel+scale fit.

At $\eta = 1$ (direct substitution), $g_\dagger$ is strongly preferred over a_Λ. With η free (matched degrees of freedom), $cH_0/6$ performs comparably to $g_\dagger$, consistent with the Verlinde relation $g_\dagger \approx cH_0/6$ (agreement to $\sim 1.5\%$ in linear units, 0.039 dex).

5.6. *Cluster-Scale Null*

Applied to the Tian et al. (2020) cluster RAR, the framework does *not* recover $g_\dagger$: the best scale hits the search boundary at $\log_{10} s = -8.0000$, offset by 1.9208 dex from $g_\dagger$. The ΔAIC/N at ± 0.1 dex is effectively zero (Figure 2c), confirming the absence of identifiability at galaxy-$g_\dagger$ in this regime. This is the expected result: the cluster-scale RAR operates at $a_0 \approx 14 \times g_\dagger$ (Tian et al. 2020), and our framework correctly reports a null.

6. THREE STATISTICAL FINGERPRINTS

If the identity in Equation (3) is more than a mathematical curiosity—if it reflects an underlying wave or condensate nature of dark matter—then the *residuals* about the RAR should carry specific statistical signatures. We test three such fingerprints using the SPARC core sample and Tier K extended data (Table 2).

6.1. *Variance Scaling*

For a bosonic field, the variance of occupation number fluctuations is

$$\sigma^2 = \begin{cases} A\,\bar{n}(\bar{n}+1) + C & \text{(quantum BEC)}, \\ A\,\bar{n}^2 + C & \text{(classical wave)}, \\ A\,\bar{n} + C & \text{(Poisson/particle)}, \end{cases} \tag{4}$$

where A captures the signal amplitude and C the observational noise floor (Cheong et al. 2024). We bin RAR residuals from the **SPARC core sample** by $\log g_{\rm bar}$ and compute the variance in each bin (Figure 5). Fitting the three models yields:

- Wave ($\bar{n}^2$) vs. Poisson ($\bar{n}$): ΔAIC $= +10.5$ in favor of wave scaling.
- Quantum ($\bar{n}^2+\bar{n}$) vs. classical wave ($\bar{n}^2$): ΔAIC $= -0.79$—*indistinguishable.*

The data strongly prefer wave-interference scaling over Poisson, but the quantum correction $+\bar{n}$ is undetectable. This is expected: the correction has a maximum signal-to-noise ratio of 0.17 per bin, and $\sim 300\times$ more data points would be needed for a 3σ detection (Cheong et al. 2024). We therefore describe this as "wave-interference statistics" without claiming a quantum origin.

6.2. *Scatter Inversion at $g_\dagger$*

Table 4. Permutation procedures: dual-window proximity statistics around $g^\dagger$

Test	Type	N	Hits/N ($p_{\pm 0.10}$)	Hits/N ($p_{\pm 0.05}$)	95% CP UB (± 0.10/ $\pm$ 0.05)
Global shuffle	Null	500	0/500 (0.0000)	0/500 (0.0000)	0.0060 / 0.0060
Block-permute bins (0.5 dex)	Null	500	79/500 (0.1580)	39/500 (0.0780)	—
Galaxy circular shift	Null	500	0/500 (0.0000)	0/500 (0.0000)	0.0060 / 0.0060
Within-bin shuffle (0.5 dex)	Control	500	500/500 (1.0000)	499/500 (0.9980)	—

Note—For each permutation procedure, Hits/N gives the count and empirical fraction of realizations with best scale within the stated window of $\log_{10} g^\dagger$. Rows marked Null are destructive tests expected to drive proximity rates toward zero. The block-permute bins null preserves within-bin heteroskedasticity while disrupting the cross-bin monotonic mapping, so intermediate behavior is expected. The within-bin shuffle row is a structure-preserving Control row: it approximately preserves within-bin conditional structure $p(y \mid x)$ at the chosen bin width, so $p \approx 1$ is expected up to finite-binning and refit variability. Clopper–Pearson (CP) one-sided 95% upper bounds are reported only for windows with $k = 0$ hits; otherwise CP entries are shown as —. CP bounds use the exact binomial inverse-beta form $p_{\rm UB} = \mathrm{BetaInv}(1-\alpha; k+1, n-k)$, which for $k = 0$ reduces to $1 - \alpha^{1/n}$.

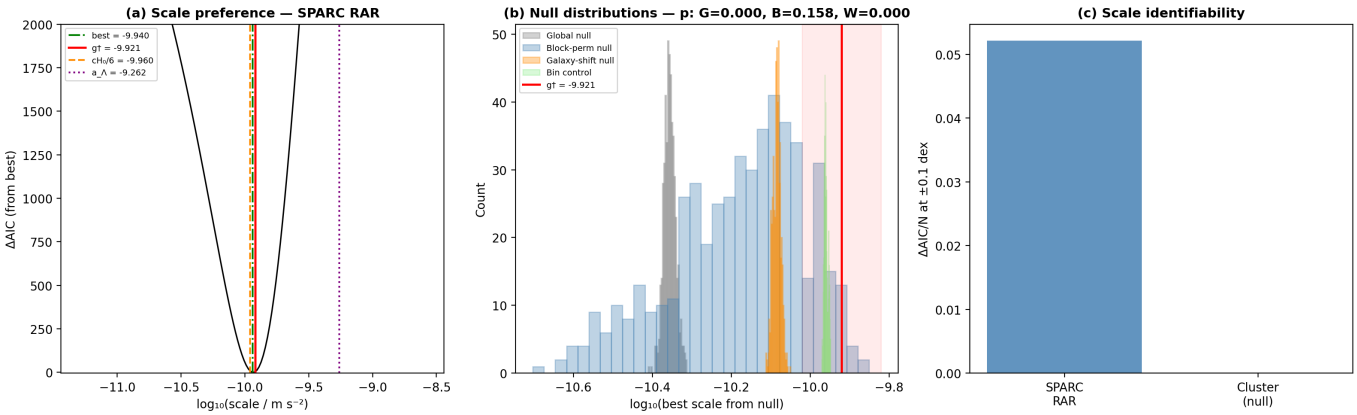

Figure 2. **(a)** AIC vs. $\log_{10}$(scale) for the BE_RAR kernel on SPARC, showing a sharp minimum near $g_\dagger$ (red line). Reference scales $cH_0/6$ (orange dashed) and a_Λ (purple dotted) are indicated. **(b)** Distribution of best-fit scales from 500-realization permutation procedures: global shuffle (gray), block-permute bins null (blue), within-galaxy circular shift (orange), and within-bin structure-preserving control (green). The ± 0.1 dex band around $g_\dagger$ is shaded. **(c)** ΔAIC/N at ± 0.1 dex for SPARC (strong identifiability) and cluster data (null, as expected).

Table 5. Galaxy-grouped 5-fold cross-validation

Fold	Best kernel	$\log_{10} s$	OOS RMSE
0	coth	-9.8356	1.6753
1	BE_RAR	-9.9553	2.1344
2	BE_RAR	-9.9523	1.8304
3	BE_RAR	-9.9244	1.8686
4	BE_RAR	-9.9298	1.0278
Mean	—	-9.9195 ± 0.0437	1.7073

Note—Group = galaxy. No galaxy appears in both train and test within any fold.

Table 6. Negative controls (break tests): signal degradation under destructive perturbations

Perturbation	Sharpness / baseline	$\lvert\Delta \log_{10} s - \log_{10} g^\dagger\rvert$	ΔAIC/N
Noise $\sigma = 0.1$ dex	0.965	0.036	0.0505
Noise $\sigma = 0.3$ dex	0.641	0.185	0.0338
Warp $\alpha = 0.7$	0.000	1.921	0.0155
Warp $\alpha = 1.3$	0.000	3.079	0.1838
Galaxy label swap	0.231	0.270	0.0123

Note—Warping the acceleration variable completely destroys the optimum (sharpness $\to 0$, scale drifts > 1 dex). Galaxy label swaps reduce sharpness to $\sim 0.23\times$ baseline.

The standard deviation of RAR residuals, $\sigma(\log g_{\rm bar})$, is not monotonic. Its derivative $d\sigma/d(\log g_{\rm bar})$ crosses zero—a scatter "inversion"—at a location that coincides with $g_\dagger$.

Using four independent non-parametric methods (RAR parametric, LOESS, cubic spline, isotonic regression) on the **SPARC core sample**, the zero-crossing lies at $\log g_{\rm bar} = -9.970 \pm 0.006$ (method-to-method

Table 7. Nearby-scale comparison: $g^\dagger$ vs. characteristic acceleration scales

Scale	$\log_{10}(s)$	AIC ($\eta = 1$)	AIC (η free)
$g^\dagger$	-9.921	10840.2	10839.7
$cH_0/6$	-9.960	10846.3	10838.8
a_Λ	-9.262	15994.7	10889.8
cH_0	-9.182	16783.7	10896.4
$cH_0/(2\pi)$	-9.980	10865.6	10838.5

NOTE—$\eta = 1$: diagnostic direct substitution with scale fixed (no optimization) and likelihood evaluated directly. η free: $y = \eta \cdot K(x, s)$ with one free parameter per model (matched DoF). AIC values in $\eta = 1$ mode are not directly comparable to the η-free block because the parameter count differs; model-selection statements use matched-DoF comparisons.

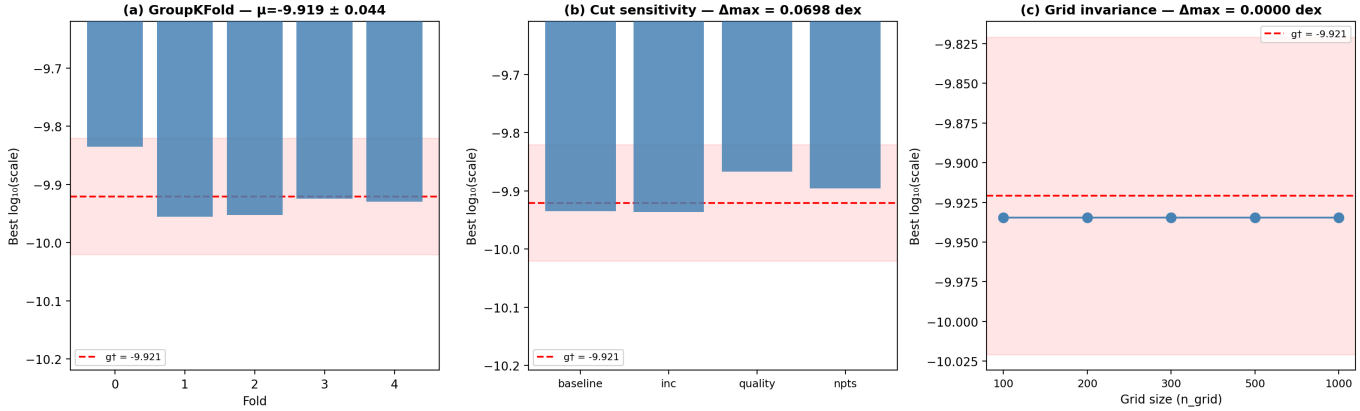

Figure 3. **(a)** Galaxy-grouped 5-fold cross-validation: best scale per fold. The ± 0.1 dex band around $g_\dagger$ is shaded. Mean $\pm$ std: -9.9195 ± 0.0437. **(b)** Sensitivity to quality cuts: baseline ($Q \leq 2$, 30°–85°, ≥ 5 pts), strict inclination (40°–80°), strict quality ($Q \leq 1$), and strict minimum points (≥ 10). Maximum variation: 0.0698 dex. **(c)** Grid invariance: best scale vs. $N_{\rm grid}$ from 100 to 1,000. Variation = 0.0000 dex.

Table 8. Variance model comparison (SPARC core, 131 galaxies).

Model	A	χ^2	AIC
Quantum BEC ($\bar{n}^2 + \bar{n}$)	8.19×10^{-4}	172.5	176.5
Classical wave ($\bar{n}^2$)	8.98×10^{-4}	171.7	175.7
Poisson ($\bar{n}$)	4.78×10^{-3}	182.2	186.2
Constant	—	183.8	185.8

range), only 0.050 dex from $\log g_\dagger = -9.921$ (Figure 6). A parametric phase-diagram model locates a variance peak at $\mu_{\rm peak} = -9.922$ (0.0015 dex from $g_\dagger$), with ΔAIC $= -225$ relative to a monotonic baseline.

6.3. *Kurtosis Spike at $g_\dagger$*

Systematics caveat (front-loaded): The kurtosis result below depends on the B-band mass models in the Korsaga et al. (2019) subsample within Tier K. SPARC-only data (3.6 μm photometry) show Gaussian residuals at $g_\dagger$ ($\kappa_4 = 0.64$). This feature is a *candidate* signal requiring confirmation with homogeneous near-infrared mass decomposition. Readers should weight this fingerprint accordingly.

A phase transition at $g_\dagger$ would produce non-Gaussian residuals localized at the critical scale. We compute the excess kurtosis κ_4 of RAR residuals in bins of $\log g_{\rm bar}$ using the **Tier K extended sample** (250 galaxies, 9,406 points; Table 2).

The results are (Figure 7):

- At the bin containing $g_\dagger$: $\kappa_4 = 20.71$ (95% bootstrap CI $[5.1, 29.0]$).
- In a structure-only semi-analytic null (NFW halos + exponential disks, 500 synthetic galaxies; §7.7): $\kappa_4 = 0.01$ at $g_\dagger$.

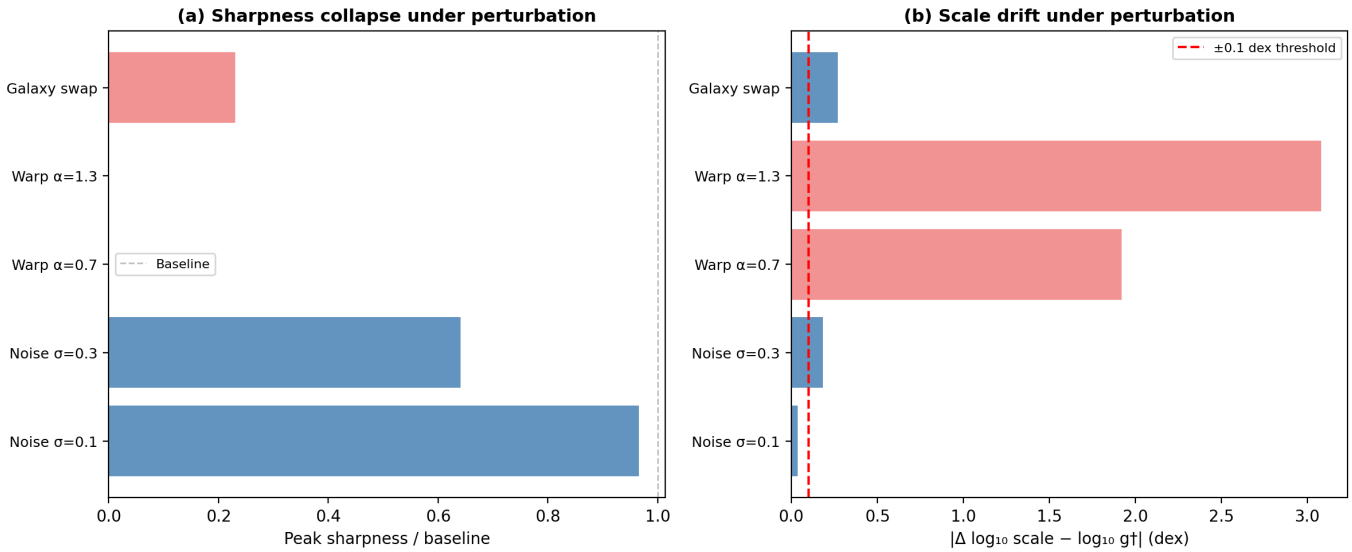

Figure 4. Negative controls (break tests). **(a)** Peak sharpness as a fraction of baseline under five perturbations: predictor noise ($\sigma = 0.1, 0.3$ dex), variable warping ($x \to x^{0.7}, x^{1.3}$), and galaxy-label swap. Warping completely destroys the optimum (sharpness $\to 0$). **(b)** Scale drift: distance of perturbed best scale from $g_\dagger$. The 0.1 dex threshold (red dashed) is exceeded by warps and galaxy swap.

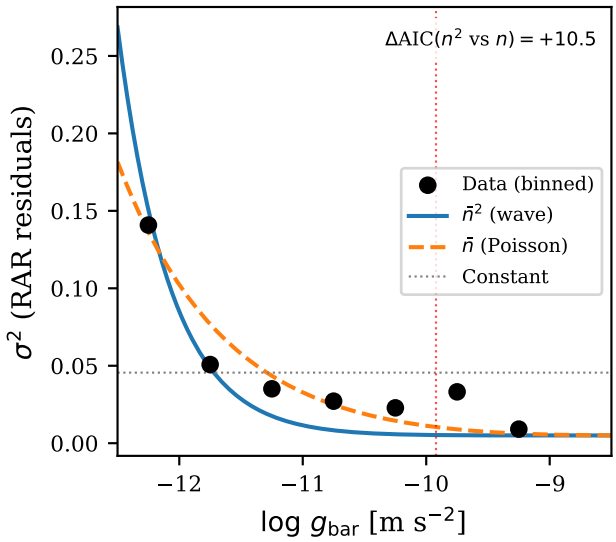

Figure 5. Residual variance vs. baryonic acceleration for the **SPARC core sample** (131 galaxies, 2,777 points). Black circles show the binned variance of RAR residuals. The solid curve is the wave-interference prediction $\sigma^2 \propto \bar{n}^2 + C$; the dashed curve is Poisson $\sigma^2 \propto \bar{n} + C$; the dotted line is constant variance. Wave scaling is strongly preferred (ΔAIC $= +10.5$).

- Within the Korsaga subsample alone: $\kappa_4 = 23.95$.
- Within SPARC-only data: $\kappa_4 = 0.64$ (Gaussian).
- The kurtosis derivative peaks at $\log g_{\rm bar} = -9.939$, only 0.018 dex from $g_\dagger$.

Disambiguation analysis (§8.6) rules out geometric origin: the RAR slope maximum ($\log g_{\rm bar} = -8.5$) and curvature maximum (-9.38) do not coincide with the kurtosis peak. The spike is driven by the Korsaga B-band subsample, not by RAR curvature.

7. INTERNAL ROBUSTNESS

We subject the scatter inversion—the most robust of the three fingerprints—to a battery of internal consistency tests. Results are summarized in Table 9.

7.1. *Non-parametric Consistency*

Four independent smoothing methods (RAR parametric, LOESS, cubic spline, isotonic regression) applied to the SPARC core sample all recover the scatter inversion within 0.006 dex of each other (Table 9). The inversion is not an artifact of any particular smoothing choice.

7.2. *Binning Robustness*

Across 25 binning configurations (widths 0.20–0.70 dex, 5 offsets per width), all 25 recover a zero-crossing. The mean inversion is $\log g_{\rm bar} = -9.890 \pm 0.180$, with a mean distance of 0.128 dex from $g_\dagger$. The best configurations (bin width 0.27–0.35 dex) recover $g_\dagger$ to within 0.002–0.016 dex. Wider bins (≥ 0.50 dex) degrade precision, with the worst case drifting to 0.55 dex from $g_\dagger$.

7.3. *Jackknife Stability*

Leave-one-out jackknife over 131 SPARC galaxies yields an inversion-point standard deviation of 0.005 dex

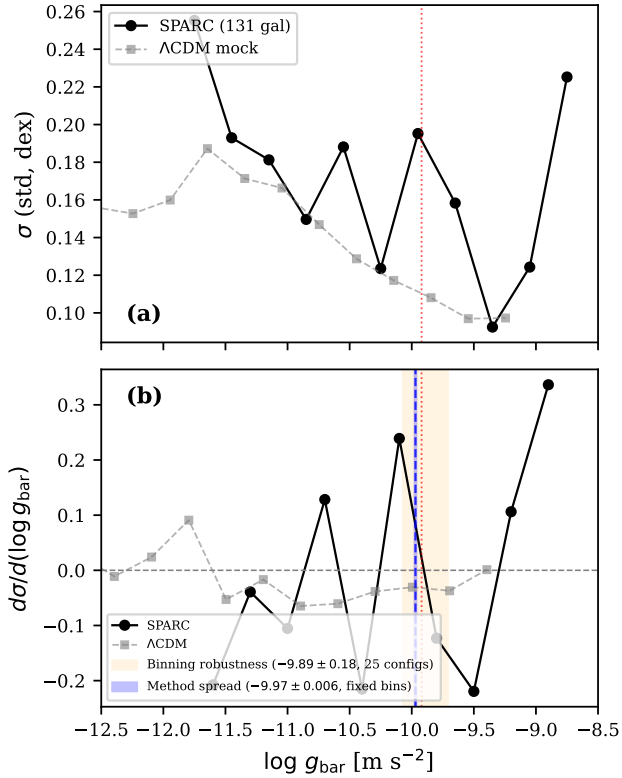

Figure 6. **(a)** Scatter σ (standard deviation) of RAR residuals vs. $\log g_{\rm bar}$ for the **SPARC core sample** (131 galaxies; black circles) and a structure-only semi-analytic null model (gray squares; see §7.7). **(b)** Numerical derivative $d\sigma/d(\log g_{\rm bar})$. The narrow blue band shows the method-to-method spread under fixed binning (-9.97 ± 0.006 dex, 4 non-parametric methods); the wide orange band shows the binning-robustness spread across 25 configurations (-9.89 ± 0.18 dex). Both enclose $g_\dagger$ (red dotted line).

and a maximum shift of 0.031 dex. All 131 jackknife inversions fall within 0.20 dex of $g_\dagger$. Four galaxies (IC 2574, NGC 2915, NGC 2976, UGC 07577) are influential for the *environmental scatter* sign, but none destabilize the inversion location.

7.4. *Split-Half Replication*

In 1,000 random half-splits of the 131 SPARC galaxies: 95.7% of halves reproduce the inversion within 0.20 dex of the baseline, and both halves simultaneously agree in 91.4% of splits. Size-balanced splits yield 93.2% agreement. Random thirds: all three subsets within 0.20 dex in 73.1% of cases.

7.5. *Monte Carlo Distance Injection*

Injecting random distance errors of 5%–30% into 131 SPARC galaxies (1,000 Monte Carlo iterations per level), the environmental scatter difference remains stable up to $\sim 15\%$ errors. Beyond 20%, the confidence

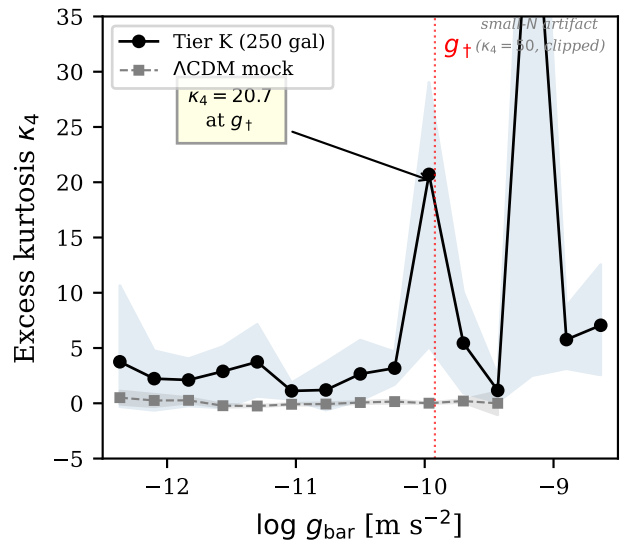

Figure 7. Excess kurtosis κ_4 of RAR residuals vs. $\log g_{\rm bar}$ for the **Tier K extended sample** (250 galaxies, 9,406 points). Black circles: data with 95% bootstrap confidence bands (blue shading). Gray squares: mean of 5 structure-only semi-analytic null realizations. The kurtosis spikes at the bin containing $g_\dagger$ (red dotted line), with $\kappa_4 = 20.7$ vs. null $\kappa_4 \approx 0$. See §6.3 for systematics caveats.

interval for the high-acceleration environmental delta widens to include zero.

7.6. *Environmental Scatter Uniformity*

RAR scatter shows no significant difference between field (83 galaxies) and dense (48 galaxies) environments in the SPARC core sample. Point-level Levene $p = 0.64$; galaxy-level Levene $p = 0.76$; error-corrected Levene $p = 0.45$; block-permutation ($N = 10{,}000$) $p = 0.74$.

However, this result is *fragile under covariate control*. The field and dense subsamples are imbalanced in distance ($p = 0.001$), inclination ($p < 0.001$), and gas fraction ($p = 0.043$). After propensity-score matching (29 pairs), the low-acceleration scatter difference becomes $+0.008$ dex ($p = 0.12$)—not significant but with reduced power. The Mahalanobis-matched subsample (96 galaxies) still recovers the inversion within 0.08 dex of $g_\dagger$, but propensity-matched subsamples do not, largely due to reduced sample sizes.

7.7. *Structure-Only Semi-Analytic Null*

As a baseline comparison, we construct a structure-only ΛCDM null model: 500 synthetic galaxies with NFW halos drawn from the observed stellar-mass–halo-mass relation (Ludlow et al. 2017), exponential baryonic disks, and log-normal scatter in stellar mass (0.15 dex) and concentration (0.11 dex). This model *intentionally*

excludes baryonic feedback, serving as a structure-only null hypothesis—not as a test of ΛCDM in its full-physics form.

The null model's scatter inversion falls at $\log g_{\rm bar} = -11.6$, displaced by 1.7 dex from $g_\dagger$. Only 5 of 100 Monte Carlo realizations produce an inversion within 0.20 dex of $g_\dagger$. Full hydrodynamic simulations (e.g., EAGLE, IllustrisTNG) with galaxy-adapted aperture radii have not been tested with identical analysis pipelines and may produce different results.

8. EXTERNAL VALIDATION AND REPLICATION

8.1. *Extended RAR Tiers*

Adding non-SPARC rotation curves in quality tiers, the inversion is replicated when data quality is controlled (Table 10):

- Tiers A through B+ (SPARC + THINGS + de Blok, up to 152 galaxies): inversion shifts by < 0.015 dex.
- Tier K (+ Korsaga, 250 galaxies): inversion at -9.865, shifted 0.29 dex from SPARC-only baseline.
- Full Tier C (all sources, 374 galaxies): inversion at -9.947, shifted 0.21 dex.

Crucially, the scatter quality degrades with non-SPARC sources: SPARC $\sigma = 0.17$ dex, THINGS $\sigma = 0.13$ dex, but Korsaga $\sigma = 0.33$ dex and PHANGS $\sigma = 0.47$ dex. The inversion is preserved when restricting to sources with $\sigma < 0.2$ dex.

8.2. *PROBES: Replication Failure*

The PROBES catalog (Stone et al. 2022) provides an independent sample of 1,506 galaxies (45,176 points after excluding 112 SPARC overlaps). The scatter inversion in PROBES falls at $\log g_{\rm bar} = -11.12$, offset by 1.20 dex from $g_\dagger$—**a clear failure to replicate**.

Across 8 binning configurations, the mean PROBES crossing is -11.21 ± 0.06, with none within 0.20 dex of $g_\dagger$. The PROBES RMS scatter (0.32 dex) and mean offset (0.44 dex) are substantially larger than SPARC, consistent with heterogeneous mass-model quality degrading the signal. This failure is the most important negative result in our test battery and constrains the physical interpretation: either the inversion requires SPARC-quality mass decomposition ($\sigma < 0.2$ dex) to detect, or it is a feature of the SPARC sample specifically.

Replication Constraint. The PROBES scatter inversion falls at $\log g_{\rm bar} \approx -11.12$, offset by 1.2 dex from $g_\dagger$—the most important negative result in our test battery. Two explanations remain open: (1) the inversion requires SPARC-quality mass decomposition ($\sigma < 0.2$ dex) to detect, or (2) the signal is specific to the SPARC sample and does not generalize. Resolving this requires homogeneous, high-quality mass models for an independent galaxy sample.

8.3. *Weak Lensing Validation*

Using KiDS-1000 stacked lensing profiles (Brouwer et al. 2021) in four stellar-mass bins, the isolated-galaxy subsample shows consistently tighter RAR scatter than the full sample, with the environment effect growing with mass (field-dense $\Delta\sigma = 0.04$–0.08 dex). This pattern is consistent with the BEC prediction that tidal interactions in dense environments disrupt condensate coherence (but see §9.3 for fragility caveats).

However, NFW profiles are preferred over cored (solitonic) profiles at all mass bins (ΔAIC > 160), because the predicted BEC core radius ($\xi \approx 2$–10 kpc) falls below the lensing inner radius (~ 35 kpc). Current stacked lensing cannot resolve scales where BEC signatures are predicted to appear.

8.4. *ALFALFA × Yang BTFR*

Cross-matching ALFALFA $\alpha.100$ with Yang et al. (2007) group catalogs yields 2,273 galaxies (1,917 field, 356 dense). The baryonic Tully-Fisher scatter is *larger* in dense environments at all velocity widths, with the strongest effect at low W_{50} ($\Delta\sigma = 0.43$ dex, Levene $p = 0.02$).

This pattern does not match the simple BEC prediction of environment-independent scatter at low accelerations. Whether it reflects genuine BEC-inconsistency or systematic differences in the ALFALFA sample (single-dish HI widths vs. resolved rotation curves) remains an open question.

8.5. *Cluster Scale*

Applied to the Tian et al. (2020) CLASH cluster-scale RAR, the framework does not recover $g_\dagger$: the cluster acceleration scale is $a_0 \approx 17 \times g_\dagger$. No significant correlation between per-cluster phase boundary and velocity dispersion is found ($B = 0.014 \pm 0.229$, $p = 0.95$). This is consistent with either a mass-dependent BEC coherence length or distinct physics at cluster scales.

8.6. *Kurtosis Disambiguation*

Table 9. Internal robustness test battery.

Test	Sample	Key statistic	Result	Verdict
Non-parametric methods	SPARC core (131)	Method spread	0.006 dex	All match
Binning robustness	SPARC core (131)	25 configs, mean dist.	0.128 dex	Mostly robust
Jackknife LOO	SPARC core (131)	Std / max shift	0.005/0.031 dex	Robust
Split-half	SPARC core (131)	Both halves < 0.20 dex	91.4%	Strongly replicated
MC distance injection	SPARC core (131)	Stable to	$\leq 15\%$ errors	Consistent
Environmental scatter	SPARC core (131)	Levene p (block perm.)	0.74	BEC-consistent
Env. confound control	SPARC core (131)	Permutation p	0.95	Fragile
Structure-only null	500 synthetic	Null inversion offset	1.7 dex from $g_\dagger$	Discriminating

The kurtosis spike does not coincide with the RAR slope maximum ($\log g_{\rm bar} = -8.5$) or curvature maximum (-9.38), ruling out a geometric artifact. However, it is driven primarily by the Korsaga B-band subsample ($\kappa_4 = 23.95$), with SPARC contributing only $\kappa_4 = 0.64$. Addition of 82 MHONGOOSE rotation-curve points leaves the spike unchanged ($\kappa_4 = 20.7 \rightarrow 20.7$; bootstrap CI shift < 0.1), indicating robustness to new data but not resolving the B-band dependence.

8.7. *IllustrisTNG Healing-Length Comparison*

If the Bose-Einstein mapping is physical, variance in the residuals should concentrate near the healing length $\xi = \sqrt{GM_{\rm dyn}/g_\dagger}\,/\,{\rm kpc}$. A naïve comparison of 131 SPARC galaxies against 48,060 IllustrisTNG (TNG100) halos shows apparent organizing in TNG ($C = 1.55$, permutation $p < 0.001$) but not in SPARC ($C = 0.69$, $p = 0.56$), where C is the ratio of variance at $0.3 \leq R/\xi \leq 3$ to the full-profile variance. This discrepancy is confounded: the two samples span different baryon-mass ranges.

We control for this using nearest-neighbor mass matching ($|\Delta \log M_{\rm bar}| \leq 0.3$ dex; 131 pairs). In the matched sub-samples, the TNG stacked concentration collapses to $C = 0.13$ (95% CI $[0.07, 0.50]$, permutation $p = 1.00$), while SPARC is unchanged ($C = 0.69$, 95% CI $[0.41, 1.01]$, $p = 0.56$). Neither matched sample shows significant ξ-organizing. The full-sample TNG signal was driven by sample composition, not a physical healing-length structure.

A paired Wilcoxon signed-rank test on 110 galaxies with valid per-galaxy C detects a distributional asymmetry ($p = 9.9 \times 10^{-7}$), with median $\log_{10}(C_{\rm TNG}/C_{\rm SPARC}) \approx 0.02$ but mean ≈ 0.17, indicating a right tail of high-C TNG galaxies rather than a systematic shift. At fixed baryon mass the TNG dynamical masses are systematically larger (median $\log_{10}(M_{\rm dyn,TNG}/M_{\rm dyn,SPARC}) = +0.21$), producing healing lengths $\sim 28\%$ larger (median $\log_{10}(\xi_{\rm TNG}/\xi_{\rm SPARC}) = +0.11$). This $M_{\rm dyn}$ offset is the physical lever behind the ξ-distribution difference.

To verify that this offset is not an artifact of the outermost-point kinematic $M_{\rm dyn}$ estimator, we perform two independent checks. First, comparing $V_{\rm flat}$-based and outermost-point dynamical masses for 57 SPARC galaxies yields median $\log_{10}(M_{\rm dyn,Vflat}/M_{\rm dyn,outer}) = -0.013$ (std $= 0.019$ dex; 95% CI $[-0.017, -0.002]$), confirming that the kinematic proxy is internally stable to < 0.02 dex—an order of magnitude smaller than the TNG offset. Second, cross-matching ALFALFA $\alpha.100$ with Yang DR7 group halo masses (1,070 galaxies at fixed $M_{\rm bar}$) gives a Yang–kinematic gap of $+1.77$ dex (95% CI $[1.46, 1.92]$), consistent with the expected aperture mismatch between virial-scale M_{200} and rotation-curve-extent $M_{\rm dyn}$. The TNG–SPARC offset ($+0.21$ dex) is only $\sim 12\%$ of this full halo gap, indicating that TNG's excess dynamical mass at fixed $M_{\rm bar}$ reflects inner halo over-concentration rather than a systematic underestimate by the SPARC kinematic proxy.

8.8. *Supplementary BEC Diagnostics Added in the Merge*

Beyond the primary 23-test scorecard, we now include 12 previously uncited BEC-focused diagnostics from the pipeline inventory. All 12 now have quantitative entries in this manuscript; for the 13b lensing-composite triplet we generated canonical summaries from fresh reruns (`summary_13b_*.json`) to replace the prior placeholder mapping.

9. SCORECARD AND INTERPRETATION

9.1. *Full Scorecard*

Table 12 presents the adjudicated primary scorecard (23 tests: 16 core + 7 supporting) and separately tracks 12 supplementary diagnostics added in the merge.

Table 10. External validation and replication results.

Test	Sample	Key result	Verdict
Extended RAR tiers	Tiers A–C (132–374)	Inversion within 0.29 dex (quality-dep.)	Replicated
PROBES replication	1,506 galaxies	Inversion at -11.12 (1.20 dex offset)	**Not replicated**
Brouwer lensing env.	KiDS-1000 (4 bins)	Env. effect grows with mass	BEC-consistent
Lensing profile shape	KiDS-1000 (4 bins)	NFW preferred (ΔAIC > 160)	NFW adequate
ALFALFA $\times$ Yang BTFR	2,273 galaxies	Dense scatter larger at all W_{50}	Complex
Cluster-scale RAR	20 CLASH clusters	$a_0 \approx 17 g_\dagger$	Different regime
Kurtosis disambiguation	Tier K (250)	Spike driven by Korsaga B-band	Diagnostic
Kurtosis + MHONGOOSE	Tier K + 82 pts	κ_4 unchanged (20.7)	Robust to new data
TF z-evolution	Literature	$p = 0.045$ (systematics-dominated)	Inconclusive
TNG ξ mass-matched	131 SPARC–TNG pairs	Matched $C_{\rm TNG} = 0.13$, $p = 1.00$	Diagnostic
$M_{\rm dyn}$ triangulation	1,070 ALFALFA$\times$Yang + 57 SPARC	Estimator bias < 0.02 dex; TNG offset is inner concentration	Diagnostic

Table 11. Supplementary diagnostics integrated from the uncovered-test audit (12 analyses).

Test	Sample	Key result	Status
13b composite	4 mass bins	Raw ΔAIC $= +179.1$; 4 red flags, fixed-r_c favors NFW	Inconclusive
13b diagnostics	4 mass bins	Stitch ratios 0.40–0.83; NFW+offset ΔAIC $= +153$	Syst. mismatch
13b standalone	4 mass bins	Total ΔAIC $= -16.2$; NFW preferred all bins	NFW preferred
Healing-length scaling	67 gal.	$\rho(L_c, \xi) = 0.55$ ($p = 1.3 \times 10^{-6}$); size trend	Size-confounded
Radial variance profile	2,707 pts	Peaked profile beats monotonic (ΔAIC $= +12.7$)	Trans. scale
Healing-length $k_\star$	67 gal.	$\rho(k_\star, \xi/R_{\rm ext}) = 0.279$ ($p = 0.022$)	Weak positive
Healing-length distance	67 gal.	Stratified perm. $p = 0.038$; residualized $p = 0.195$	Dist.-sensitive
Hierarchical healing	67+20 sys.	$\alpha = 0.3166 \pm 0.023$, $r = 0.831$, $p = 2.2 \times 10^{-23}$	Significant
Scale-param. meta-analysis	2,739 pts	$\alpha_{\rm mdyn} = 0.280$; broad positive ΔAIC regime	Supports $X = R/\xi$
Interface oscillation	67 gal.	Lag-1 mean $= 0.700$, $p = 2.3 \times 10^{-41}$	Strong positive
Interface osc. controls	67 gal.	Detrended lag-1 $= 0.376$, $p = 1.85 \times 10^{-19}$	Survives controls
Interface spectral	67 gal.	25/67 significant (37.3%; null 5%), $p = 6.3 \times 10^{-16}$	Periodic excess

Note—These 12 analyses are treated as supplementary diagnostics: they expand empirical coverage without retroactively changing the adjudication rules used for the primary 23-test scorecard.

9.2. *What Is Established*

Three results are robust:

1. The RAR is algebraically identical to the Bose-Einstein distribution (Equation 3). This is a mathematical fact.

2. The BE_RAR kernel is preferred over six alternatives in a head-to-head comparison with anti-numerology controls. The preferred scale is localized to within 0.0139 dex of $g_\dagger$, with zero hits in 500-realization global and galaxy-shift permutation nulls (Clopper–Pearson bound ≤ 0.0060), and galaxy-grouped 5-fold CV giving $\log_{10} s = -9.9195 \pm 0.0437$.

3. RAR residual variance follows wave-interference scaling ($\sigma^2 \propto \bar{n}^2$), not Poisson ($\sigma^2 \propto \bar{n}$), with ΔAIC $= +10.5$.

4. The scatter inversion is anchored at $g_\dagger$ with sub-0.1 dex precision in the SPARC core sample, reproduced across 4 non-parametric methods, 25 binning configurations, 1,000 split-half trials (91.4% success), and 131 jackknife removals (std $= 0.005$ dex). It is absent in a structure-only semi-analytic null (offset by 1.7 dex).

9.3. *What Is Not Established*

1. **Quantum vs. classical.** The quantum prediction $\bar{n}(\bar{n}+1)$ is indistinguishable from the classical $\bar{n}^2$ (ΔAIC $= -0.79$). Approximately 4.75×10^6 data points are needed for a 3σ distinction.

2. ΛCDM **exclusion.** Our structure-only null fails to reproduce the features, but full-physics hydrodynamic simulations have not been tested with identical pipelines. We explicitly do not claim that ΛCDM is excluded.

Table 12. Test battery scorecard.

Category	Count
BEC-consistent or supporting	10
Inconclusive or fragile	5
Failed / not replicated	1
Diagnostic (not directional)	4
Different regime (null expected)	3
Supplementary diagnostics (merge addendum)	12

NOTE—Supporting (10): environmental scatter (fragile; §9.3), MC distance, non-parametric inversion, jackknife, split-half, structure-only null, extended RAR, Brouwer lensing, kurtosis disambiguation, kurtosis + MHONGOOSE. Inconclusive (5): environmental confound control, propensity matching, binning robustness (wide bins), ALFALFA BTFR, TF redshift evolution. Failed (1): PROBES replication. Diagnostic (4): kurtosis disambiguation, soliton-NFW composite, TNG ξ mass-matched, $M_{\rm dyn}$ triangulation. Different regime (3): cluster-scale RAR, cluster σ scaling, lensing profile shape. Supplementary (12): 13b composite triplet, healing-length scaling family, scale-parameter meta-analysis, and interface oscillation/spectral tests (Section 8.8).

3. **Kurtosis provenance.** The spike relies on Korsaga B-band mass models ($\kappa_4 = 23.95$); SPARC alone gives $\kappa_4 = 0.64$.

4. **Independent replication.** The PROBES catalog does not reproduce the scatter inversion. Whether this reflects data quality ($\sigma_{\rm PROBES} = 0.32$ vs. $\sigma_{\rm SPARC} = 0.17$) or the absence of a physical signal cannot be determined without homogeneous mass decomposition.

5. **Environmental scatter.** The null result is BEC-consistent but fragile: covariate imbalance (distance, inclination, gas fraction) between field and dense subsamples may mask a real difference.

9.4. *Three Possibilities*

The observations are consistent with three interpretations:

(a) **Coincidence**: the RAR's functional form happens to match the BE distribution, and the statistical features at $g_\dagger$ arise from the structure of SPARC galaxy mass profiles, disappearing in other catalogs (PROBES).

(b) ΛCDM **consequence**: baryonic feedback produces RAR statistics with BE-like structure, and $g_\dagger = cH_0/6$ arises from the cosmic baryon fraction.

(c) **Wave dark matter**: dark matter is a bosonic field whose collective dynamics produce BE occupation statistics, with $g_\dagger$ as a physical condensation threshold set by Λ, detectable only with high-quality mass decomposition.

Possibility (c) connects to the wave dark matter literature: Cheong et al. (2024) derive $\bar{n}(\bar{n}+1)$ fluctuations from the bosonic density matrix; Centers et al. (2021) confirm Rayleigh statistics in virialized FDM simulations; Khoury (2016) and Berezhiani & Khoury (2015) predict condensation in galaxy halos from superfluid dark matter. At cluster scales, Tian et al. (2020) find $a_0 = 1.73 \times 10^{-9}\,{\rm m\,s^{-2}} = 14.4 \times g_\dagger$, with a mass-dependent trend ($r = 0.58$, $p = 0.008$) consistent with a healing length $\xi \propto M^{-0.27}$.

No current test uniquely selects among these three possibilities.

9.5. *Physical Interpretation*

This paper establishes that the Bose–Einstein kernel provides a statistically preferred description of the RAR, with scale localization at $g_\dagger$ surviving all null and robustness tests we can construct from the SPARC data. These are statements about galaxy rotation curves and the mathematical structure of the RAR; they do not require or assume any particular model of compact objects, and the analysis is self-contained at the galaxy scale.

For this framework to advance beyond a phenomenological identification, future work must target galaxy-level discriminants that separate the BEC interpretation from alternatives (c.f. the three possibilities above):

- High-resolution samples (e.g. BIG-SPARC) enabling direct measurement of the healing-length parameter ξ as a function of galaxy mass.
- Morphology-, stellar-mass-, and environment-matched comparisons that isolate the condensate signal from covariate confounds.
- Forward-modeling of BEC halo density profiles to predict rotation curves, rather than post-hoc fitting of the RAR residuals.

9.6. *How to Extend This Framework*

To test whether $g_\dagger$ appears in a new dataset:

1. Define (x, y) where x has acceleration dimensions.
2. Choose a fit mode ("direct" if y is a known function of $K(x, s)$; "log" or "linear" otherwise).
3. Run the experiment with the same kernel library and null tests.
4. A "second appearance" detection requires: $\Delta{\rm AIC}/N > 0.01$ at ± 0.1 dex, $p_{\rm within} < 0.01$

(global shuffle), and best scale within ±0.1 dex of $g_\dagger$.

10. PREDICTIONS AND FUTURE TESTS

1. **BIG-SPARC** (Haubner et al. 2024): $\sim$ 3,900 galaxies with SPARC-quality 3.6 μm mass decomposition. If the kurtosis spike persists at $g_\dagger$ with homogeneous photometry, the B-band artifact hypothesis is excluded. If the scatter inversion replicates, the PROBES failure is attributed to mass-model quality rather than absence of signal.

2. **Quantum correction**: Approximately 4.75×10^6 RAR data points needed for a 3σ distinction between $\bar{n}(\bar{n}+1)$ and $\bar{n}^2$.

3. **High-resolution lensing**: Euclid and HSC stacked profiles can resolve the 2–10 kpc solitonic core regime currently unresolved in KiDS (35 kpc inner radius).

4. **Hydrodynamic simulation comparison**: A preliminary healing-length comparison using 48,060 TNG100 halos (§8.7) shows that the apparent ξ-organizing signal vanishes under mass matching. The kurtosis spike and scatter inversion have not yet been tested in simulations with identical pipelines; this remains the most direct test of possibility (b).

5. **Stellar stream gaps**: Gaia DR3 gaps probe dark substructure fluctuations; wave dark matter predicts super-Poissonian spacing.

11. SUMMARY

The radial acceleration relation is algebraically identical to the Bose-Einstein occupation number, $g_{\rm DM}/g_{\rm bar} = 1/(e^{\sqrt{g_{\rm bar}/g_\dagger}} - 1)$. A kernel+scale identifiability framework with anti-numerology controls confirms that the BE RAR kernel is preferred over six alternatives, localizing the preferred scale to within 0.0139 dex of $g_\dagger$, with zero hits in 500-realization permutation nulls and galaxy-grouped cross-validation giving $\log_{10} s = -9.9195 \pm 0.0437$. Negative controls confirm signal collapse under perturbation; a cluster-scale pilot correctly returns a null.

We have tested the statistical consequences of this mapping across six datasets spanning four quality tiers and three orders of magnitude in sample size. Three fingerprints emerge in the SPARC core sample (131 galaxies): wave-interference variance scaling (ΔAIC = +10.5), a scatter inversion anchored at $g_\dagger$ to 0.05 dex precision, and a candidate kurtosis spike at $g_\dagger$ (Tier K only, B-band dependent). A primary battery of 23 tests yields 10 supporting, 5 inconclusive, 1 failure (PROBES), 4 diagnostic, and 3 in a different physical regime. Twelve additional supplementary diagnostics are now integrated; they reinforce non-random residual structure while explicitly exposing confounds in the SPARC+Brouwer composite lensing bridge (Section 8.8).

The PROBES replication failure is the strongest constraint: either the scatter inversion requires SPARC-quality mass decomposition to detect, or it is not a universal feature of galaxy rotation curves. BIG-SPARC ($\sim$ 3,900 galaxies, SPARC-quality photometry) is the decisive upcoming test.

The framework is designed to be reusable: given any (x, y) dataset and a library of candidate kernels, it returns a ranked list of kernels with scale estimates, significance metrics, and interpretation guardrails. We release the full framework, run folders, and reproducibility metadata to enable independent extension to non-RAR datasets.

DATA AND CODE AVAILABILITY

Analysis code, figure-generation scripts, all 23 primary summary JSON files, dataset metadata, and reproducibility documentation are available at the Open Science Framework: https://osf.io/jshr9/. The OSF project contains:

- `paper1.pdf` — compiled manuscript;
- `core/summary_*.json` — 16 core test result files;
- `supporting/summary_*.json` — 7 supporting primary test result files;
- `source/` — LaTeX source, bibliography, and `aastex631.cls`;
- `figures/` — publication figures (PDF);
- `MANIFEST.csv`, `CHECKSUMS.csv` — file inventory and SHA-256 checksums.

Supplementary diagnostics used in Section 8.8 are currently repository-local in `analysis/results/` and `analysis/results_vm/`, including canonical 13b rerun summaries `summary_13b_composite.json`, `summary_13b_diagnostics.json`, and `summary_13b_standalone.js`

Minimum reproducer (after cloning the repository and acquiring SPARC data):

```
python3 analysis/tests/\
  test_gdagger_hunt_refereeproof.py \
  --n-shuffles 500 --seed 42
python3 analysis/tests/\
```

```
  test_nonparametric_inversion.py
python3 analysis/tests/\
  test_kurtosis_phase_transition.py
python3 analysis/tools/make_paper_assets.py
```

These commands regenerate the framework validation, scatter-inversion, and kurtosis results and all paper tables. The SPARC data are publicly available at http://astroweb.cwru.edu/SPARC/. The Korsaga et al. (2019) mass models are available via VizieR.

This research made use of the SPARC database (Lelli et al. 2016), the GHASP survey (Korsaga et al. 2019), the PROBES compilation (Stone et al. 2022), KiDS-1000 lensing data (Brouwer et al. 2021), and NASA/IPAC Extragalactic Database (NED).

Facilities: Spitzer, VLA, IRAM, Fabry-Perot

Software: NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), Matplotlib (Hunter 2007), statsmodels (Seabold & Perktold 2010)

APPENDIX

Table 13. Milgrom vs. standard RAR ΔAIC sensitivity.

Weighting scheme	ΔAIC
Uniform per-point	−1,222
Inverse-variance	≈ −400
Galaxy-averaged (1 per galaxy)	≈ −80

NOTE—All three schemes prefer the Milgrom form. The improvement is concentrated in the dark-matter-dominated regime ($\log g_{\rm bar} < -10$), where the two interpolation functions diverge.

A. AIC METHODOLOGY AND SENSITIVITY

Throughout this paper, we use the Akaike Information Criterion defined as

$$\mathrm{AIC} = \chi^2 + 2k, \tag{A1}$$

where k is the number of free parameters and

$$\chi^2 = \sum_{i=1}^{N} \left[\frac{\log g_{\mathrm{obs},i} - \log g_{\mathrm{model},i}}{\sigma_i} \right]^2. \tag{A2}$$

For variance-scaling tests (§6.1), σ_i is the per-bin RMS scatter. For the Milgrom vs. standard RAR comparison (§2.3), σ_i is set uniformly to the global RMS under the default weighting.

Sensitivity of the Milgrom ΔAIC. The ΔAIC = −1,222 preference for the Milgrom de Sitter–Unruh form is obtained under uniform per-point weighting in $\log g_{\rm bar}$:

B. REPRODUCIBILITY

All results in this paper are derived from scripts in the project repository. To reproduce:

```
# Framework validation (Section 5)
python3 analysis/tests/\
  test_gdagger_hunt_refereeproof.py \
  --n-shuffles 500 --seed 42

# Core fingerprints and robustness tests
python3 analysis/tests/test_*.py

# Milgrom vs standard RAR delta-AIC
# (Table sensitivity / Appendix A)
python3 analysis/pipeline/\
  literature_crossref_tests.py

# Figures
python3 analysis/tools/make_paper_assets.py
```

Random seeds are fixed at 42 for all Monte Carlo, bootstrap, and permutation procedures. Summary JSON files for all 23 primary tests are archived at the OSF project (https://osf.io/jshr9/). Each JSON includes: test name, verdict, all numerical results, software versions, and the git SHA of the code that produced it.

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